

Module 08 Lab: Bias-Variance Tradeoff — Group 1 Reflection Journal

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Course: Intro to Machine Learning 1371

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Dana Bolden - Before doing this lab, I thought that if a model got almost everything right during training, that meant it was a good model. After going through the exercises, I realized that's not always true. I learned that a model can do really well with the data it has already seen but still perform poorly when it's given new data. That helped me understand why it's important to test models with different datasets instead of just looking at the training results. This lab changed the way I think about measuring how well a machine learning model actually works.

Jonathan Aldana - Exploring the Bias-Variance Tradeoff provides a hands-on understanding of how model complexity directly impacts performance across training and validation datasets. By generating polynomial models and plotting their learning curves, I was able to visualize the practical difference between high bias and high variance rather than relying solely on theory. This experience directly helped me answer Question 2 regarding the underfitting model. Observing the Degree 1 learning curve firsthand allowed me to see how the training score drops quickly while the cross-validation score flattens out at a similarly low level. Seeing both scores converge to a low overall performance value with no noticeable gap proved that when a model has high bias, it is too simple to capture the underlying data pattern. This visual outcome made it clear why collecting additional training samples cannot solve underfitting, directly guiding my explanation for Question 2

Astrid Lorenzana - Whilst working through previous labs, a term that would show up quite a few times was 'overfitting'. Before learning more about it, I just assumed the term stood for exceeding the standards and going over what was expected. For example, when one is "overqualified" for a position at a job. However, it is a concept that shows that our model only memorized the data it was training with, it did not learn a pattern. Understanding this is important because a model that displays output based on what it has memorized, when met with an unfamiliar input (that remains relevant to the data given) it will not know where to assign it, leading to it being unreliable.